Matrices in Geometry - 2.4.7

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Question

Find the projection of the vector $\mathbf{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ on the vector $\mathbf{b} = \hat{i} + 2\hat{j} + \hat{k}$.

Given Information

Let the vectors be represented as $\mathbf{A}$, $\mathbf{B}$ respectively. Given by

$$\mathbf{A} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \tag{1}$$

Solution

Mathematically, the projection of the vector $\mathbf{A}$ on the vector $\mathbf{B}$ is given by some vector $\mathbf{D}$,

$$\mathbf{D} = k\mathbf{B}, \text{ such that } (\mathbf{A} - \mathbf{D})^{\mathsf{T}}\mathbf{D} = 0 \tag{2}$$

Continuation

yielding,

$$(\mathbf{A} - k\mathbf{B})^{\mathsf{T}}\mathbf{B} = 0 \tag{3}$$

or,

$$k = \frac{\mathbf{A}^{\mathsf{T}}\mathbf{B}}{\|\mathbf{B}\|^2} \implies \mathbf{D} = \frac{\mathbf{A}^{\mathsf{T}}\mathbf{B}}{\|\mathbf{B}\|^2}\mathbf{B} \tag{4}$$

Continuation

On substituting the values,

$$\mathbf{D} = \frac{\left(\begin{pmatrix}2\\3\\2\end{pmatrix}\right)^{\mathsf{T}} \begin{pmatrix}1\\2\\1\end{pmatrix}}{\left\|\begin{pmatrix}1\\2\\1\end{pmatrix}\right\|^2} \begin{pmatrix}1\\2\\1\end{pmatrix} \tag{5}$$

Final Answer

On calculation, this gives us,

$$\mathbf{D} = \begin{pmatrix} 5/3 \\ 10/3 \\ 5/3 \end{pmatrix} \tag{6}$$

C code

```
#include <stdio.h>

void vector_projection(double a[3], double b[3],
double proj[3]) {
double b_norm_sq = b[0]*b[0] + b[1]*b[1] + b[2]*b[2];
double dot_product = a[0]*b[0] + a[1]*b[1] + a[2]*b[2]

    double scalar = dot_product / b_norm_sq;

    proj[0] = scalar * b[0];
    proj[1] = scalar * b[1];
    proj[2] = scalar * b[2];
}
```

Python code

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Define vectors
a = np.array([2, 3, 2])
b = np.array([1, 2, 1])

# Calculate projection of a on b
b_norm = b / np.linalg.norm(b)
proj_a_on_b = np.dot(a, b_norm) * b_norm

# Plotting
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
```

Python code

```
# Plot vector a
ax.quiver(0, 0, 0, a[0], a[1], a[2], color='r',
label='Vector a')

# Plot vector b
ax.quiver(0, 0, 0, b[0], b[1], b[2], color='b',
label='Vector b')

# Plot projection of a on b
ax.quiver(0, 0, 0,proj_a_on_b[0],proj_a_on_b[1],
proj_a_on_b[2],color='g',label='Projection of a on b')

# Setting the plot limits
max_val = max(np.linalg.norm(a),np.linalg.norm(b))+1
ax.set_xlim([0, max_val])
ax.set_ylim([0, max_val])
```

Python code

```
ax.set_zlim([0, max_val])
ax.set_xlabel('X axis')
ax.set_ylabel('Y axis')
ax.set_zlabel('Z axis')
ax.legend()
plt.title('Vector Projection: a on b')
plt.show()
```

Plot

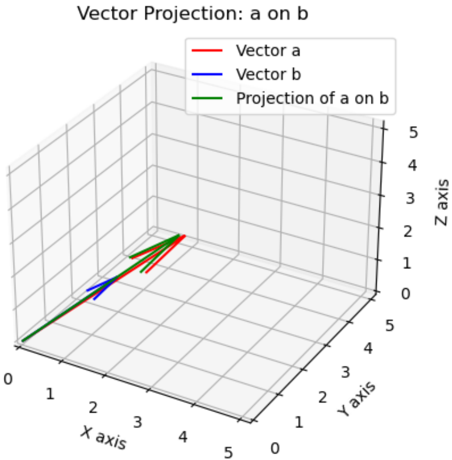

Figure: 3D Plot